JAIDEV EDUCATION SOCIETY'S

J D COLLEGE OF ENGINEERING AND MANAGEMENT

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(An Autonomous Institute, with NAAC "A" Grade)

Affiliated to DBATU, RTMNU & MSBTE Mumbai

Department of CSE

"A Place to Learn; A Chance to Grow"

2024-25 (Even Sem)

VISION	MISSION
To be a recognized as an Excellent Innovative Engineering Department through Academic Programme which develops leaders in Education and Research in Computer Science.	1. To create self-learning environment by facilitating leadership qualities, team-spirit and ethical responsibilities. 2. To Strengthen department-industry collaboration and interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Subject Name: Design & Analysis of Algorithms **Sem/Branch :- 4th /CSE** **Subject Code :- CS4T002**

Date of Display: 27/03/2025

Date of submission: 03/04/2025

Assignment

1. Write the significance of amortized analysis and explain all its three methods **CO 1/ L1** **2 M**
2. Explain different Asymptotic notations. **CO 1/ L2** **2 M**
3. Explain Analysis of Binary Search for following elements and calculate average number of comparisons for successful and unsuccessful Search.

   -12, 23, 31, 45, 56, 78, 90, 103, 113, 126, 157. **CO 2/ L2** **2 M**
4. Write Algorithm for Quick Sort. Sort the following array using Quick Sort. **CO 2/ L1** **2 M**

   40, 20, 10, 80, 60, 50, 7, 30, 100
5. Find the optimal solution to the fractional knapsack. If the knapsack capacity W = 60 kg.

Item	I_1	I_2	I_3	I_4	I_5
Weight	5	10	15	22	25
Cost	30	40	45	77	90

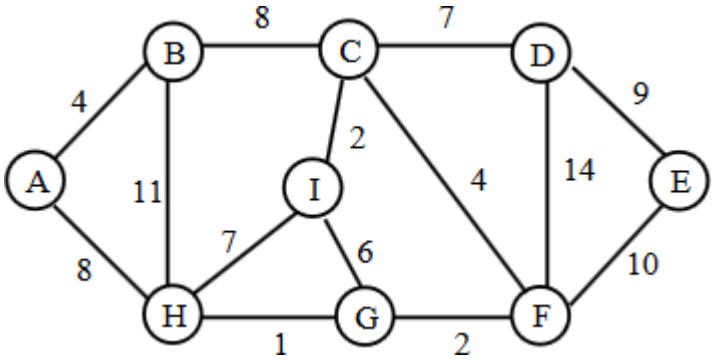
CO 3/ L2 **2 M**

6. Define optimal merge pattern? Implement optimal binary merge pattern for ten files whose length are

   38, 42, 22, 15, 94, 63, 101, 43, 13, 21 **CO 3/ L2** **2 M**

7. Write algorithm to find out minimum cost spanning tree for the following graph implement PRIMS

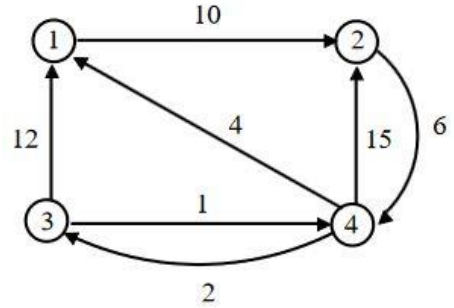
algorithm. **CO 3/ L1** **2M**



8. Write Huffman code algorithm. Also find optimal Huffman code for following set of frequencies.

a:20, b:15, c:5, d:25, e:35, f:11 **CO 3/ L1** **2 M**

9. Write an algorithm for Floyd Warshall & Solve following. (7 M) **CO4/L1** **2M**



10. Define Longest Common Subsequence? Find the LCS length for given strings. **CO1/L1** **2M**

String X = b, q, d, r, c, v, e, f, g, h

String Y = a, b, c, v, d, e, f, g, h

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Subject Incharge	Academic Incharge	HoD